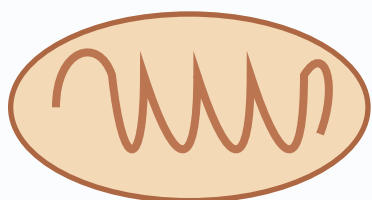


CELLULAR RESPIRATION

Add-on cutouts • Use with your photosynthesis molecule cards

Cut on dashed lines. Each picture is a symbol, not a measured quantity.

MITOCHONDRION



Most stages occur here

Simplified organelle diagram

ATP



Energy transferred to ATP

Usable energy for cell work

Battery = symbol for energy storage

HEAT



Energy released as heat

Transferred to the surroundings

Build the respiration model: Reuse all 19 molecule cards. Arrange glucose and 6 oxygen molecules on the reactants side; arrange 6 carbon dioxide and 6 water molecules on the products side.

Matter equation:



Add the symbols: Place the mitochondrion above the arrow. Place ATP and heat beside the products to show energy transfer. Set the sun and chlorophyll cards aside. The energy cards are annotations; do not include them in the atom count.

Teacher sentence: “Cellular respiration transfers some of the chemical energy in glucose to ATP, which cells use to do work, and releases some energy as heat.”

Accuracy note: Respiration begins in the cytoplasm; most later stages in plant and animal cells occur in mitochondria. ATP is made from ADP and phosphate, which this simplified model does not show. One ATP card does not mean that only one ATP molecule is produced.

Check: Both sides contain 6 carbon, 12 hydrogen, and 18 oxygen atoms. Plants carry out cellular respiration too.